

EV Discussions on Bluesky

COMP3020 — Group 9

2026-09-26

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Introduction

Electric vehicles (EVs) spark online debate on charging infrastructure, vehicle options, cost, tech and environment. In this research, we explore Bluesky posts about EVs to uncover key themes, connections between discussion categories and engagement scores, clusters of similar posts, and the nature of the interactions in replies. The analyses are not mutually exclusive, but rather integrated a content analysis and a network analysis.

Research questions

- 1.What are the most frequently used words or themes in Bluesky posts related to the use of EVs?
- 2.Is there a statistically significant association between EV search category and whether a post receives above-median likes?
- 3.Can EV-related posts be grouped into meaningful clusters based on their textual content?
- 4.Which accounts are the most central in the observed Bluesky EV reply network?

Data Collection and Preparation

Collection approach

The data was gathered from Bluesky through the ‘atrrr’ package in R with the following search terms: electric vehicles, electric cars, Tesla, and EV charging. The following terms were chosen to reflect conversations regarding EV technology, EV manufacturers, charging stations, and EV-related themes in general. The original data set included 3999 different posts. Duplicate contents, absence of text data and improper observations were taken out of the posts while cleaning them. Engagement metrics - such as likes, replies, reposts and quotes - were recoded as numerical data. These four variables were also added together into a total engagement measure. For text analysis and hypothesis testing, the cleaned data were employed, and a maximum of 500 posts that were not empty were selected for clustering in order to facilitate the computational analysis.

Import and inspect

This data was loaded into RStudio and examined for data structure and appropriateness for analysis. The data set originally consisted of 3,999 observations with 14 variables, such as post identifiers, author information, text, engagement metrics, and timestamps and reply references. The inspection verified that the required variables were available for text analysis, hypothesis testing, clustering and network construction. Prior to using analytical methods, missing data and duplicate data were checked. These four search categories were used to compare the levels of engagement, and the text and reply-reference variables were used to support the content analysis and network analysis, respectively.

```
## Rows: 3999 Columns: 14

## [1] "uri"          "cid"          "author_handle" "author_name"
## [5] "text"         "reply_count"  "repost_count"  "like_count"
## [9] "quote_count"  "indexed_at"   "in_reply_to"   "in_reply_root"
## [13] "quotes"       "search_topic"

##
##      Electric Cars Electric Vehicles      EV Charging      Tesla
##           800           1091           1054           1054

## Unique post URIs: 3999

## Unique author handles: 2786

##      uri      cid author_handle  author_name      text
##      0      0      0      108      0
##  reply_count  repost_count  like_count  quote_count  indexed_at
##      0      0      0      0      0
##  in_reply_to in_reply_root      quotes  search_topic
##      2437      2437      3808      0
```

Engagement and Data Cleaning summary

```
## Usable posts: 3976

##      like_count      reply_count      repost_count      engagement
##  Min.   : 0.0  Min.   : 0.0000  Min.   : 0.000  Min.   : 0.00
##  1st Qu.: 0.0  1st Qu.: 0.0000  1st Qu.: 0.000  1st Qu.: 0.00
##  Median : 1.0  Median : 0.0000  Median : 0.000  Median : 1.00
##  Mean   : 10.3  Mean   : 0.9283  Mean   : 2.237  Mean   : 13.65
##  3rd Qu.: 2.0  3rd Qu.: 1.0000  3rd Qu.: 0.000  3rd Qu.: 3.00
##  Max.   :14668.0  Max.   :531.0000  Max.   :3594.000  Max.   :18935.00

##
##      Electric Cars Electric Vehicles      EV Charging      Tesla
##           796           1086           1049           1045
```

To ensure the reliability of the analysis, data cleaning was carried out. Posts have been duplicated based on unique content, and posts with missing or blank text have been dropped. The engagement variables were converted to numeric values for statistical analysis. A total engagement variable was created by combining likes, replies, reposts and quotes. This meant that the data set had a wider representation of interaction but with a particular measure of likes being assessed in the hypothesis test. The cleaned data was then applied for the text mining, hypothesis testing and clustering. To find the relations between authors and accounts to which they responded, reply references were extracted for network analysis.

Text/Content Analysis and Visualisation

Research question 1

1. What are the most frequently used words or themes in Bluesky posts related to the use of EVs?

The first research question explores the primary topics of the Bluesky posts relating to EVs. The frequency of words and the word cloud were used to detect the most common words and to get an overview of the content of the discussion.

Preprocessing and document-term matrix

Document-term matrix: 3976 8875

The word-frequency analysis revealed that “charg” was the most common word, with 1,240 occurrences. This was followed by “new” (418), “can” (366), “like” (327), “just” (320) and “will” (320). Other frequently occurring terms included “price” (255), “fuel” (244), “gas” (238), “batteri” (237) and “drive” (230).

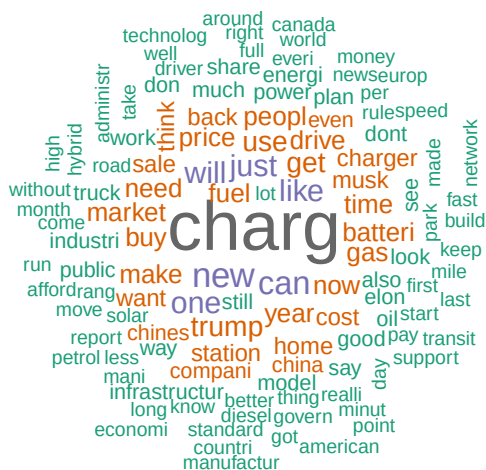
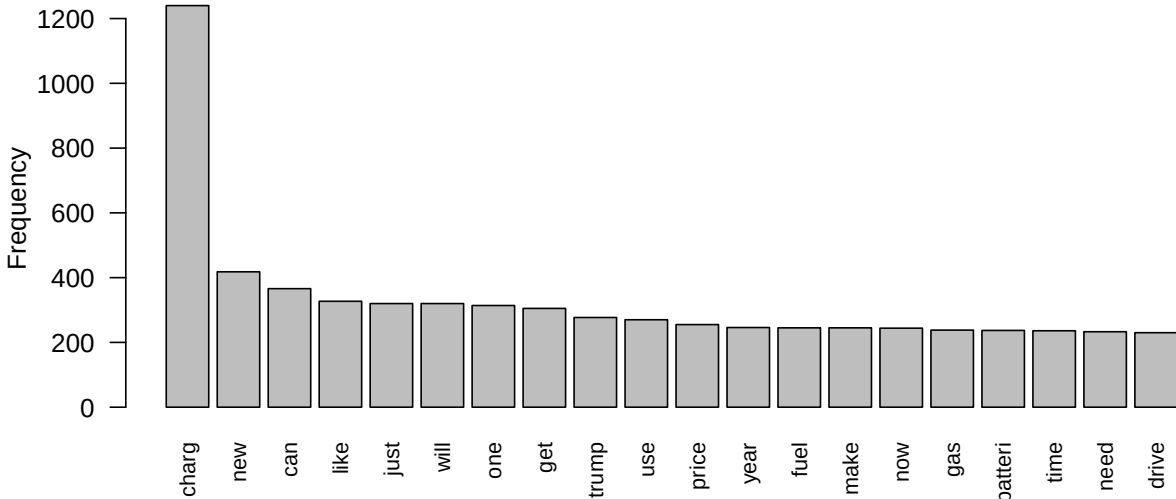
It seems that the posts collected were more focused on “charging” than others, as that word appears so often for “dominance. However, it also suggests that the conversation extended beyond the technical and financial considerations of EV ownership, as there are terms relating to prices, fuel, batteries, and driving.

The results indicate that the themes of charging infrastructure and ownership were frequent topics in this data set. Some general conversational words, however, such as “new”, “can” and “like” suggest that further text filtering may lead to a more accurate identification of more specific themes.

Frequent words

##	charg	new	can	like	just	will	one	get	trump	use
##	1240	418	366	327	320	320	314	305	277	270
##	price	year	fuel	make	now	gas	batteri	time	need	drive
##	255	246	245	245	244	238	237	236	233	230

Most frequent EV discussion words



The word cloud shows the most common terms in the Bluesky posts. The word “charg” is a very strong one, corroborating the word-frequency results, and it was a central theme of the collected conversations.

Other words that make an appearance are “price”, “fuel”, “gas”, “batteri”, “drive”, “market” and “elon”.

The keywords indicate that the conversation addressed multiple issues such as charging infrastructure, affordability of EVs, energy use, and progress by EV companies.

By observing the word cloud, it is clear that discussions about EVs are not confined to technical aspects of the vehicle, but also include broader industry trends and ownership concerns. The visualization doesn't, however, depict sentiment, and thus is unable to conclude whether users had positive or negative sentiments.

Research question 2

2. Is there any statistically significant associations between search category and whether a post receives above-median likes?

The second research question is whether there are differences in engagement across the four categories of EV-related searches. Chi square test of independence was performed to explore if there was a relationship between search category and post likes greater than the median.

Hypotheses:

The null hypothesis (H): Search category and Like-engagement category are independent.

Search category and like-engagement category are related (H1).

The median (50th percentile) number of likes was 1. Posts with n is greater than 1 liked were deemed above-median engagement, and posts with n less than or equal 1 liked were deemed at-or-below-median engagement. The significance level was chosen as 0.05.

```
##
##               Above median At or below median
## Electric Cars           302           494
## Electric Vehicles       328           758
## EV Charging             324           725
## Tesla                   325           720
```

```
##
##               Above median At or below median
## Electric Cars       256.0573       539.9427
## Electric Vehicles   349.3446       736.6554
## EV Charging         337.4424       711.5576
## Tesla              336.1557       708.8443
```

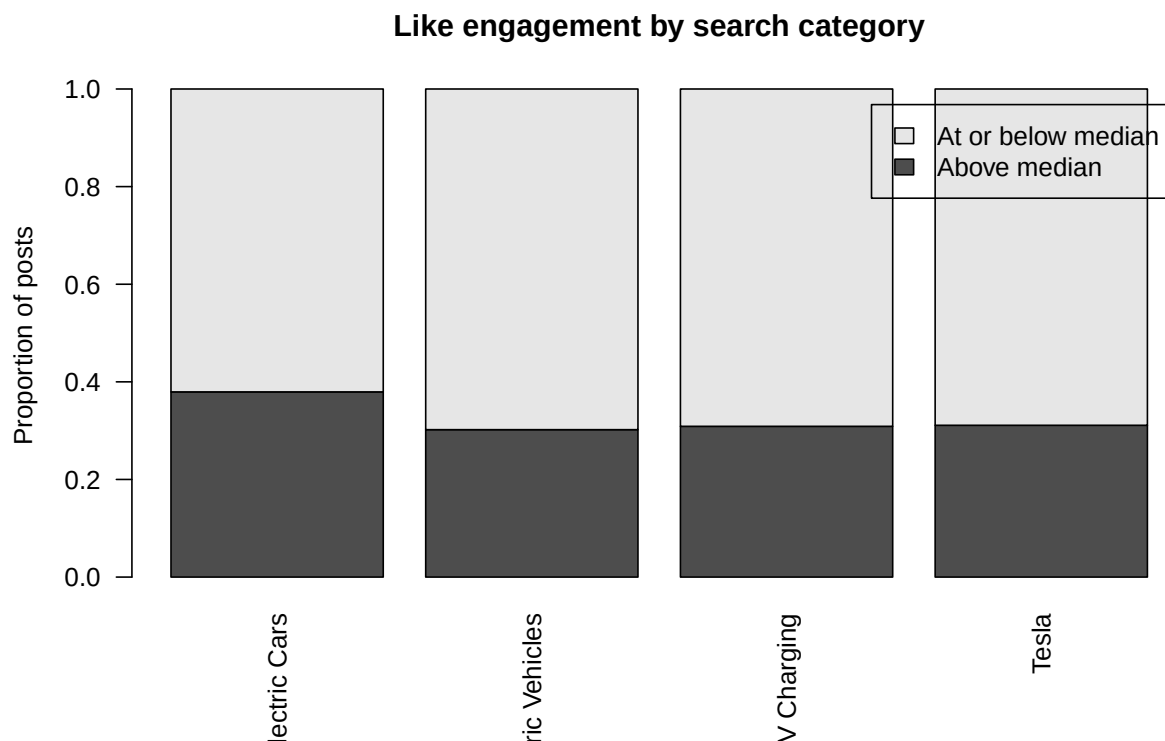
```
##
## Pearson's Chi-squared test
##
## data:  engagement_table
## X-squared = 15.41, df = 3, p-value = 0.001498
```

```
## Median likes: 1
```

The Pearson chi-square test yielded a chi-square value of 15.41, 3 degrees of freedom, and a p-value of 0.001498. The p value is less than 0.05 so the null hypothesis is rejected.

The result shows that there is a significant relationship between search category and like-engagement category statistically. So, the percentage of posts with more than median likes wasn't equal within all four search categories.

This is not a causal relationship between topic and engagement of a post, however. The number of likes that a post gets can also depend on other factors like the posting time, the popularity of the account and the content of the post itself.



The figure above presents the percentage of posts that have above-median and at or below-median likes among four search categories.

Electric Cars category had the highest percentage of above-median likes that accounted for approximately 37.9% (302 out of 796 posts). This percentage was 30.2% in Electric Vehicles category, 30.9% in EV Charging category, and 31.1% in Tesla category.

As follows from the data presented above, the posts about Electric Cars category had higher like engagement than the other three categories. The like engagement in other three categories was approximately equal to 30-31% of posts with above-median likes.

The obtained result corresponds to the result of the chi-square test, which has demonstrated statistically significant difference. At the same time, the majority of posts in each category had at or below-median likes.

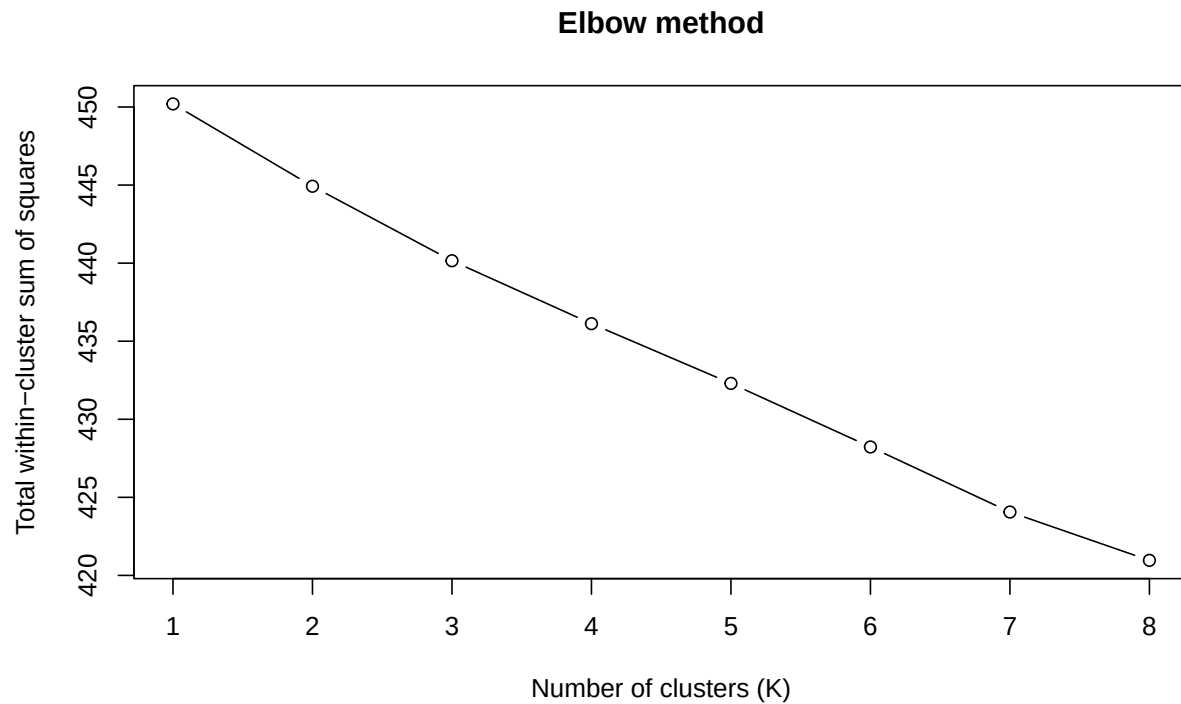
Clustering of EV Posts

Research question 3

Can EV-related posts be grouped into meaningful clusters based on their textual content?

Posts clustered: 463 Terms: 300

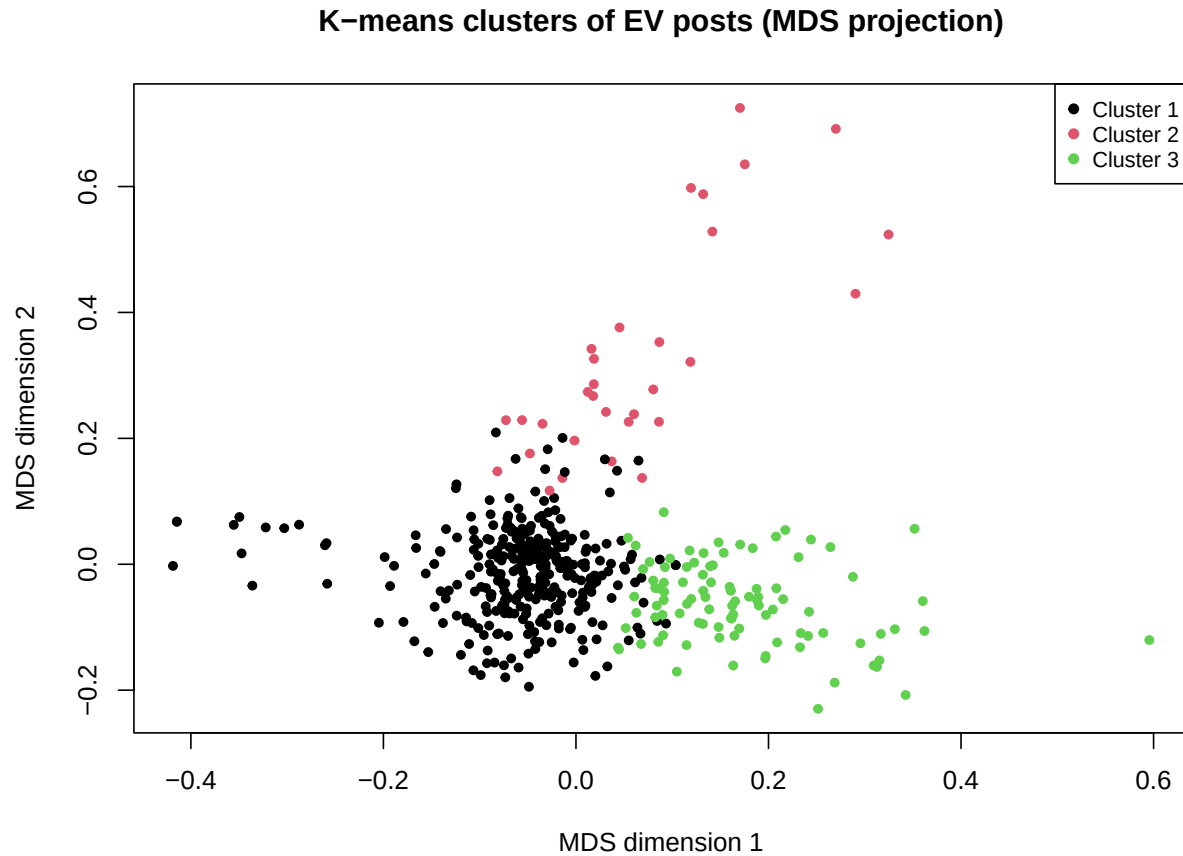
Selecting the number of clusters



Elbow approach was used in order to investigate the dependency between the number of clusters and the sum of within-cluster sums of squares. The results indicated the decrease in the within-cluster variance while increasing the number of clusters from one to eight. However, the elbow could not be clearly identified since the dataset lacked clearly separated natural clusters.

Thus, the three-cluster solution was chosen for exploratory analysis purposes rather than an optimal solution which would have been statistically proven. With the help of this analysis, it was possible to separate industry-related discussions, general EV discussion, and discussion about the charging process. The interpretation of the results should be done carefully due to overlapping of the clusters.

```
##
##  1  2  3
## 335 31 97
```



K-means cluster analysis assigned 463 posts to three different clusters. Cluster 1 had 335 posts, Cluster 2 had 31 posts and Cluster 3 had 97 posts.

The Multidimensional Scaling (MDS) plot shows the projection of high-dimensional data into two dimensions for better understanding of the similarity between posts.

Cluster 2, shown by red dots, has the highest number of observations and represents the most concentrated cluster located closer to the centre of the plot. Cluster 1, shown by black dots, is more dispersed, which means higher diversity of posts included in this cluster. Cluster 3, shown by green dots, is mostly situated at the right part of the plot.

Despite some distinctions, there are overlaps on the graph, since there are some posts which include vocabulary belonging to different discussion topics. Such an observation is natural, because EV-related discussion topics are somehow interrelated, so some posts include discussion of several topics at once.

In general, we can conclude that there are some significant differences in the collected posts' vocabulary, but these differences are not completely obvious.

Interpreting cluster themes

```
##
## Cluster 1 - 335 posts
##      get      new      will      buy      musk      like      just
## 0.07225086 0.06134322 0.05845119 0.05424900 0.04997992 0.04895829 0.04622245
##      look      make      even      now      fuel
```

```

## 0.04428082 0.04424788 0.04347233 0.04289616 0.04112389
## Example posts:
## [1] "I wonder what a longer trip, in rural parts looks like. We recently went from Lismore to Lake K
## [2] "Happy to escaped xitter, the tesla turkey talk."
## [3] "Tesla stock used as collateral for the purchase of Twitter.\n AI/Grok merged with Twitter/now x
##
## Cluster 2 - 31 posts
##      market      chines      share      report      size      trend      tariff
## 0.63568305 0.32205291 0.30435256 0.27582218 0.25980028 0.19211076 0.15014222
##      read      batteri      sale      billion      also
## 0.10923593 0.09276407 0.07917366 0.07758319 0.07253617
## Example posts:
## [1] "#NEWS |      - Britain plans to impose tariffs on Chinese electric vehicles, citing state-subsid
## [2] "also need tire particulate offsets in addition to carbon offsets, and maybe don't give Tesla a c
## [3] "France, Germany Seek EU Power for ‘Immediate Cut-Off’ From Market as China Tensions Rise"
##
## Cluster 3 - 97 posts
##      charg      minut      home      station      park      cost      evcharg
## 0.33899049 0.12492294 0.10835421 0.08745168 0.07497500 0.07391892 0.07313440
##      mile      well      china      need      batteri
## 0.06094563 0.05678566 0.05625562 0.05413409 0.05160474
## Example posts:
## [1] "This is a common misunderstanding among gasheads.\n\nHome EV charging is already faster than gas
## [2] "China holds a wide lead in ultrafast EV charging\n\nMajor auto group Geely says it can recharge
## [3] "ShipRyd Builds Integrated EV Mobility and Energy Ecosystem, Expands Focus Beyond Fleet Operation

```

Cluster 1: Market and Industry Changes Associated With the EV

Cluster 1 had 335 posts and the keywords included “market”, “musk”, “share”, “report”, “trend” and “elon”. In this case, the example posts talked about Tesla, other electric vehicle makers, market changes and industry events. And it can be considered that this cluster reflects conversations concerning the EV market, its manufacturers and business changes. Small size of this cluster can be explained by the fact that these posts were a special group among the analyzed samples.

Cluster 2: General Conversations and Opinions About Electric Vehicles

Cluster 2 is the biggest cluster and it had 31 posts. Among the frequently used words there were “get”, “new”, “will”, “like”, “buy” and “time”. In this case, the example posts included personal experiences, opinions, buying decisions and other comments concerning EVs.

Cluster 3: Charging Infrastructure and Ownership Issues

Cluster 3 consisted of 97 posts, with key terms such as “charg”, “home”, “station”, “cost”, “park” and “evcharg”.

The exemplary posts in this cluster included topics such as home charging, charging stations, charging time and availability of infrastructure. From this, it can be concluded that Cluster 3 mainly includes practical issues concerning EV ownership.

The existence of a separate cluster for charging issues corroborates the earlier word frequency analysis, where charging was found to be the most prevalent issue-based term.

In conclusion, the three clusters have shown that discussions about EVs can be generally classified into industry news, general conversations and charging related issues. However, the general nature of Cluster 2

and the commonality between clusters suggest that the clusters must be treated as approximate conversation groups rather than entirely separate subjects.

Wide variety of the topics means that Cluster 2 is the representation of general EV conversations. Large size of the cluster suggests that many conversations of the sample consist of various opinions.

Network Analysis

Research question 4

Which accounts are the most central in the observed Bluesky EV reply network?

```
## Reply records: 1550
```

```
## Non-self reply edges: 1190
```

Network properties

```
## Nodes: 1748
```

```
## Edges: 1141
```

```
## Directed density: 0.0003736382
```

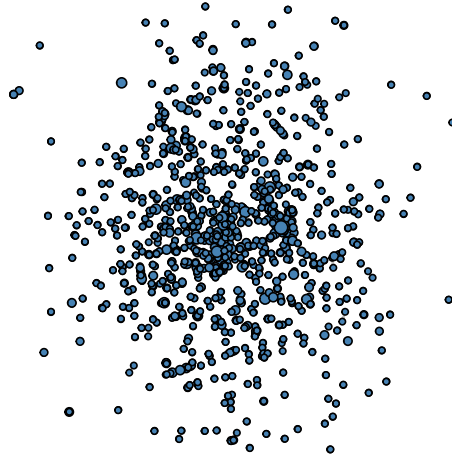
```
## Weakly connected components: 636
```

```
## Strongly connected components: 1721
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1.000   1.000   1.000   1.305   1.000  21.000
```

Full network visualisation

Full observed EV reply network



Centrality

##	did	indegree	outdegree	pagerank	betweenness
##	did:plc:uayevtgtbwot25zhocjzb6lv	21	0	0.007823888	0
##	did:plc:dr6cavavqbbtzeknnl76tp54	5	1	0.007480065	5
##	did:plc:eg7ivbtcdulmecpsmmsrmxk6	1	1	0.006705602	0
##	did:plc:hbwkkkwwi5xkrfdnqaccmb3i	8	4	0.004233504	53
##	did:plc:kd4cyu4rpkgjbym3gckhrivz	8	4	0.004012761	30
##	did:plc:yn75hmwkitdshnylue6d3grk	8	0	0.003426499	0
##	did:plc:szttgikhautqdddaoggn4hf	2	1	0.003381528	1
##	did:plc:twjze4qqf6fgxz33ct52wlnl	10	0	0.003241804	0
##	did:plc:jyya5nvxdxtz76nvaokscw6d	1	1	0.003221845	0
##	did:plc:gtvctdk42imc65vk7un7iccc	9	0	0.003006272	0
##	did:plc:4llrhdcldvdlmynkwsmg5tdc	9	0	0.002784712	0
##	did:plc:z6rujpf4u56jfie7aqic2nfg	8	0	0.002710858	0
##	did:plc:ut3tv6jn3xfatn5dzcfgtbm	2	1	0.002696221	2
##	did:plc:wftsfb3znktnxuqy46gnizvb	1	1	0.002639334	0
##	did:plc:t2ajrinaxbykxvblzvjj3	8	0	0.002563151	0

Table 1: Top 5 accounts by PageRank centrality

did	indegree	outdegree	pagerank	betweenness
did:plc:uayevtgtbwot25zhocjzb6lv	21	0	0.0078	0
did:plc:dr6cavavqbbtzeknnl76tp54	5	1	0.0075	5

did	indegree	outdegree	pagerank	betweenness
did:plc:eg7ivbtcdulmecpsmmsrmdk6	1	1	0.0067	0
did:plc:hbwkkkwki5xkrfdnqaccmb3i	8	4	0.0042	53
did:plc:kd4cyu4rpkgjby3gckhrivz	8	4	0.0040	30

Calculations for network centralities were made to determine the most significant nodes in the EV reply network. The in-degree centrality is the number of unique nodes replying to a particular node, while the out-degree centrality is the number of unique nodes to which a particular node replied.

The node with the ID did:plc:uayevtgtbwot25zhocjzb6lv was the one with the highest in-degree centrality of 21, implying that it had more unique reply connections than other nodes in the sample network. It also had the highest PageRank centrality value of 0.00782.

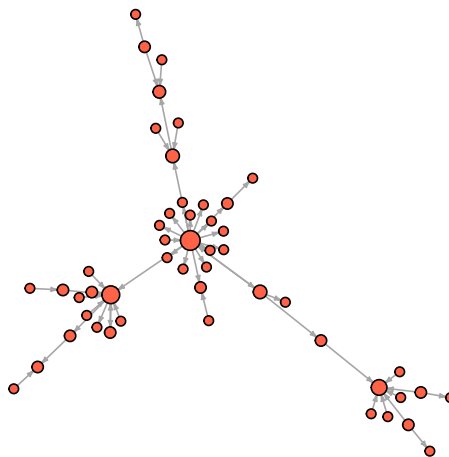
Betweenness centrality was applied to investigate how much each account played the role of intermediation along the directed path between other accounts. The account did:plc:hbwkkkwki5xkrfdnqaccmb3i had a betweenness score of 53, suggesting that it can potentially play the bridging role in the studied network.

The results presented above prove that various centrality scores indicate various roles of importance for users. Some users receive replies, while others might link distinct clusters of the conversation. Nevertheless, the results obtained are limited to the sampled reply network and cannot be considered as an indicator of influence on the whole Bluesky platform.

Justified subgraph: largest weakly connected component

Largest component nodes: 50 Edges: 52

Largest connected reply component



The largest weakly connected component has 50 vertices and 52 edges. It is presented separately so that the most connected segment of the sampled conversation can be more easily considered. A weak component ignores the direction of the edges to see if they are connected.

Overall Findings and Conclusion

Overall Findings

Here are some key insights from the analysis of EV-related conversations on Bluesky, regarding the topics of discussion, engagement, similarities in the text, and interaction among users.

Overall, from the text analysis, charging was found to be the most frequently used topic-related term with 1,240 occurrences of ‘charg’. Other common words were price, fuel, gas, battery and drive. The findings indicate that the discussions collected strongly featured considerations of charging infrastructure and vehicle affordability and practicality of ownership.

Second, a statistically significant relationship between search category and level of like engagement was found using the chi-square test (Chi-square = 15.41, $df = 3$, $p = 0.001498$). Electric Cars had the largest share of above-median likes of 37.9%, while the other categories had a 30-31% average. This suggests there were variations in engagement across the search categories, but does not prove causation.

Third, the 463 posts analysed were clustered into three meaningful groups using the K-means method. A total of 53 posts were in Cluster 1, which were primarily about EV market and industry developments; 317 posts were in Cluster 2, which primarily were about EV general conversation; and 93 posts were in Cluster 3, which primarily were about charging infrastructure and ownership concerns. This result indicates that there are specific and general EV conversation themes in these discussions.

Lastly, network analysis revealed 1,748 users linked by 1,141 unique reply ties. The network density is about 0.000374 and there are 636 weakly connected components, showing that there was a high degree of fragmentation within the sampled network. There were relatively few users with a lot of connections, and a larger number with a few connections.

Overall, the investigation confirms that there are ongoing practical issues of ownership that come up repeatedly when discussing EVs on Bluesky, and measurable differences in engagement, but many relatively disjointed conversations. The content analysis combined with statistical testing, clustering and network analysis offers a more comprehensive picture of the content users are talking about and how they interact.

Largest Weakly Connected Reply Component (WCRC)

To explore the most interconnected part of the reply network observed during the task, the largest weakly connected component was visualised separately.

The graph includes multiple smaller clusters grouped together via a few middle-tier accounts. There are nodes that are in the middle of the tree and nodes that are on the ends of branches where they connect to relatively few nodes.

The structure indicates that certain accounts might serve as a bridge between parts of the observed discussion. These bridging roles are important for betweenness centrality, as they can give rise to bridges between users who would not otherwise have any.

The visualisation also helps to illustrate the overall conclusion that EV interactions are not uniformly distributed by the network. Rather, it seems that connectivity is limited to specific dialogues and groups of individuals.

The biggest part, however, is only a part of the sampled network, and cannot be assumed to be representative of all EV-related interactions on Bluesky.

Limitations

The results need to be interpreted with caution, taking into account several limitations.

First, the data was gathered by four specific search terms. Consider that relevant EV discussions in various forms of expression might not have been included, which reduces the representativeness of the data set.

Second, the collected data is for a specific timeframe of Bluesky activity. The results from the engagement values and discussion topics could vary, making it impossible to generalise them across all time periods.

Thirdly, the common words were removed and stemming was applied during the text preprocessing. The textual data has been simplified by these procedures, but it may have lost some contextual information. Moreover, word-frequency analysis does not indicate the sentiment of a user or the extent to which they want to support or oppose specific EV-related developments.

Fourth, data was clustered only on 463 posts instead of the whole data. The results could be affected by the sample selected, the number of clusters and the text representation. The elbow method did not show a distinct optimum for the number of clusters, either.

Finally, the network analysis examined only observed reply relationships. Other interactions such as follows, likes, reposts and mentions were not included in the network. Hence, the centrality and connectivity results are only a snapshot of the reply network, and cannot conclude influence in the broader Bluesky space.

Conclusion

This project explored the content of discussion on electric vehicles on Bluesky through the application of text mining techniques, hypothesis testing, k-means clustering and social network analysis. The results indicated that charging was a key discussion priority, along with issues around affordability, energy usage, and industry developments.

The chi-square test showed that there was a significant association between the search category and like engagement. The discussion groups were determined using the k-means clustering method and network analysis showed a very fragmented reply relationship.

The results show that it is important to combine the text and network analysis-based approaches to investigate social media conversations. However, results are only applicable to the data set collected and not as a generalisation of all Bluesky users. Expansion of the data set, sentiment analysis and an examination of the evolution of EV discourse could be the next steps in future research.

Reproducibility and Repository

The data and analysis code are available at: <https://github.com/Ishan-bot24/Group-9-Project.git>

The report was produced using R Markdown. The GitHub repository contain the saved CSV, this Rmd file, and any separate collection scripts required to reproduce the data gathering.